

SWAMI

The 9
0
5 Project



KUSH
YEGNASWAMI[®] 



Kush Yegnaswami

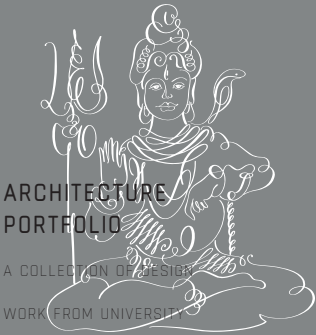
250 Queens Quay W
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ONTARIO, CAN

kushyegnaswami@gmail.com

SWAMI

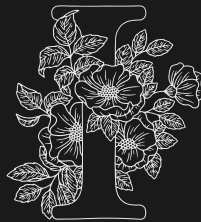
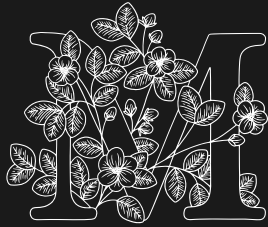
2
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2
S E L E C T E D W O R K S



ARCHITECTURE
PORTFOLIO

A COLLECTION OF DESIGN

WORK FROM UNIVERSITY



STUDI8



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S E L E C T E D W O R K S

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SWAMI



PERSONAL INFORMATION

Kush Yegnaswami

4 June 1998

Born in Calgary, Alberta

Living in Toronto, CAN

Interests:

Architectural Design

Video Production

Music Production

Business

Marketing

PROFICIENCY



Revit
Rhinoceros
Grasshopper



Sketchup
Illustrator
AutoCAD



Photoshop
Lightroom
Indesign



Logic
Final Cut
Premiere



Word
Excel
PowerPoint

SKILLS

I can create and present projects on all the above mentioned software tools. Please reach out to me if you have a specific project in mind.

www.kushswami.com / kushyegnaswami@gmail.com

Kush Yegnaswami

DESIGNER

Offering professional design services, video editing, creative product marketing, music production, and audio mixing/mastering.

EDUCATION

White Oaks Secondary School 2012-2016
Oakville
Garth Webb Secondary School

University of Toronto 2016-2020
Bachelor of Architectural Studies TORONTO
<https://www.utoronto.ca/>

EXPERIENCE

2017
TED Conferences - India MUMBAI
Researcher
<https://www.ted.com/playlists/606>

2018-2019
WSP TORONTO
Corporate Video Content Creator
<https://www.wsp.com/en-CA>

2022-Present
Circle NEW YORK (REMOTE)
First Sales Development Representative Hire
<https://circle.so/>

After receiving 24M dollars in a Series A funding round, Circle is focused on bringing their product to market and increasing revenue.

ABOUT ME BUSINESS

I hope to combine my interests for content creation and business and start my own company.

SWAMI Media

I joined University of Toronto in September 2016. In addition to designing, I love creating music. I play 3 instruments - violin, saxophone, and guitar. I enjoy producing my music, and that has gotten me familiar with piano and drum programming. I create videos of me performing original music and remixes, and I post them to Instagram, Tik Tok and Youtube.

@KUSH_SWAMI

Volume I

The 905 Project

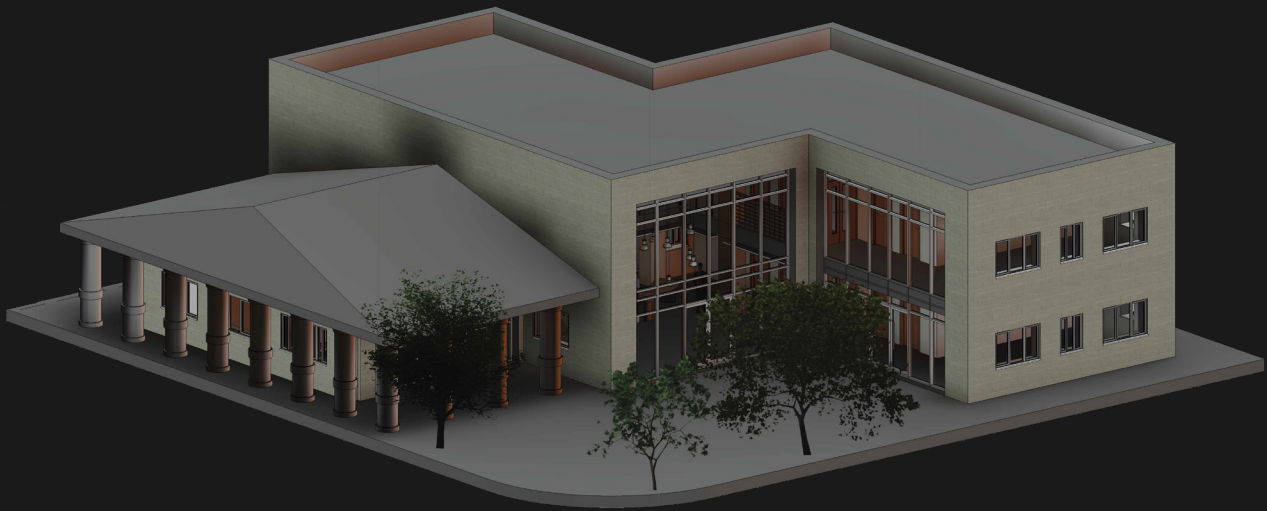
SWAMI

01

URBAN DESIGN

OFFICE BUILDING

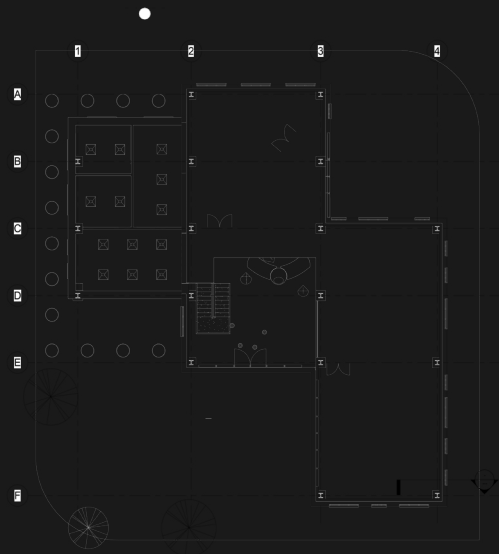
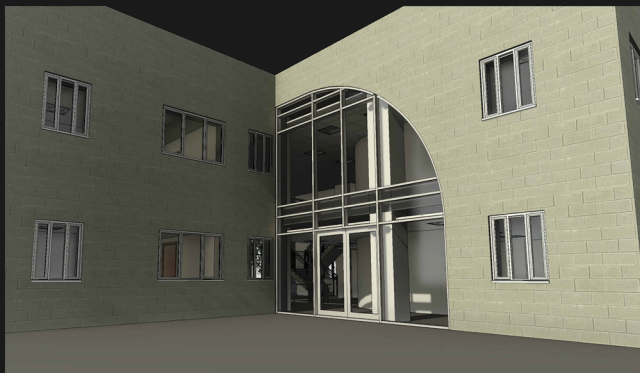
A PERSONAL PROJECT



INFORMATION

While I enjoyed studying at UofT, I felt that the design degree that I did didn't allow me to work on many 'real life' architectural projects. Our projects were mainly conceptual design exercises. Although I enjoyed those projects and they did help me to explore my creative interests, this is a personal project I took on that helped to understand Revit better and utilize the software in a real world type of scenario.

Using online tutorials, I got aquanited with Revit, the software industry standard, and I learned how to utilize the software to design a real world building.



PROJECT 01 — 2021



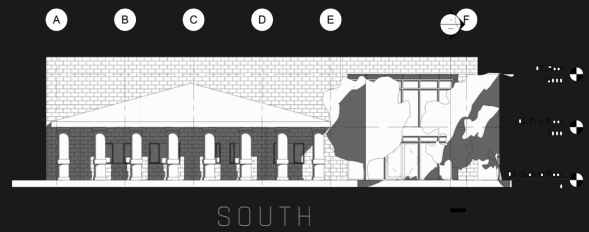
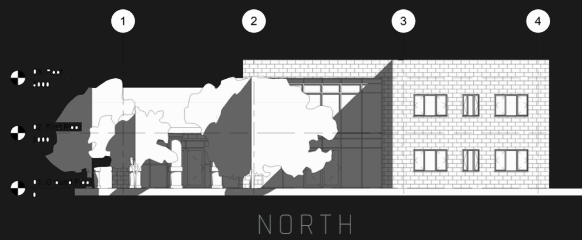
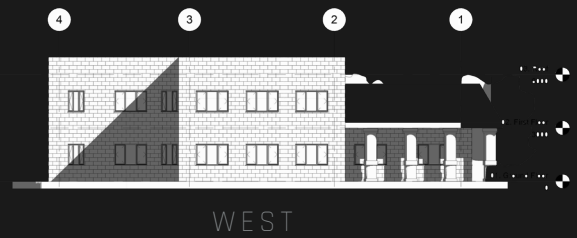
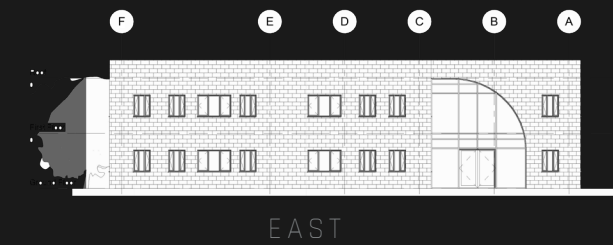
Volume I

905

0
Page 8

A modern office building should both look beautiful as well as function as an efficient space for professionals. If it lacks beauty, inhabitants will lack inspiration for the work that they do.

URBAN ARCHITECTURE



PERSPECTIVE

Inspired by some of the office buildings I visit in downtown Toronto, the lobby of this imagined building is modern in its aesthetic presentation yet traditional in its layout. I aimed to create a professional yet cozy feeling when you walk through the doors.

URBAN ARCHITECTURE

WALKTHROUGH

A walk-through of this building in Revit makes it evident right away that everything has been strategically placed in order to maximize for efficiency.

PRACTICAL DESIGN

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9

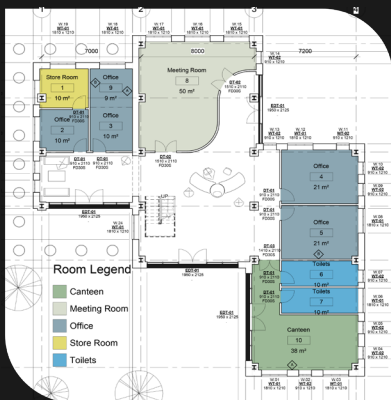
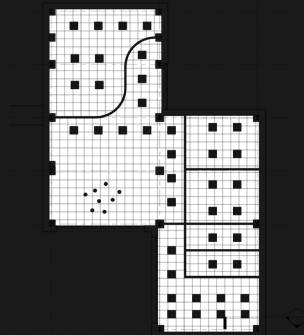
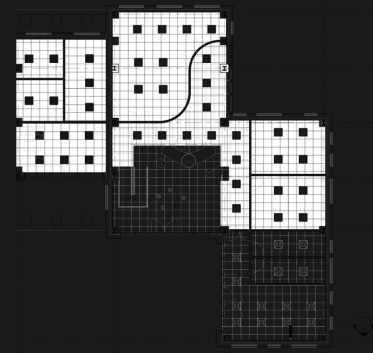
SUSTAINABLE DEVELOPMENT: TWO SCHOOLS OF THOUGHT

One school of thought when approaching urban design is to develop urban areas in harmony with natural systems. Currently, suburban developments remain largely low density and automobile focused, which does significantly reduce environmental load. However, giving special care to the preservation of water and green space will create an even more lasting influence for suburban developments. In this project I took special care to implement green space.

Another school of thought is to increase urban density in order to reduce reliance on automobiles. This means making streets as pedestrian friendly as possible. On a regional level that means creating multifunctional centers that integrate seamlessly with public transit. In a perfect world, I would want this office building to be right next to a subway station so that employees are encouraged to take public transportation.

This building was designed help employees feel comfortable and inspired.

This building is designed with detail end to end. Below are the ceiling details for each floor, as well as a room legend.



10
ROOMS
BUILT IN
2021
SIZE
110 sqm

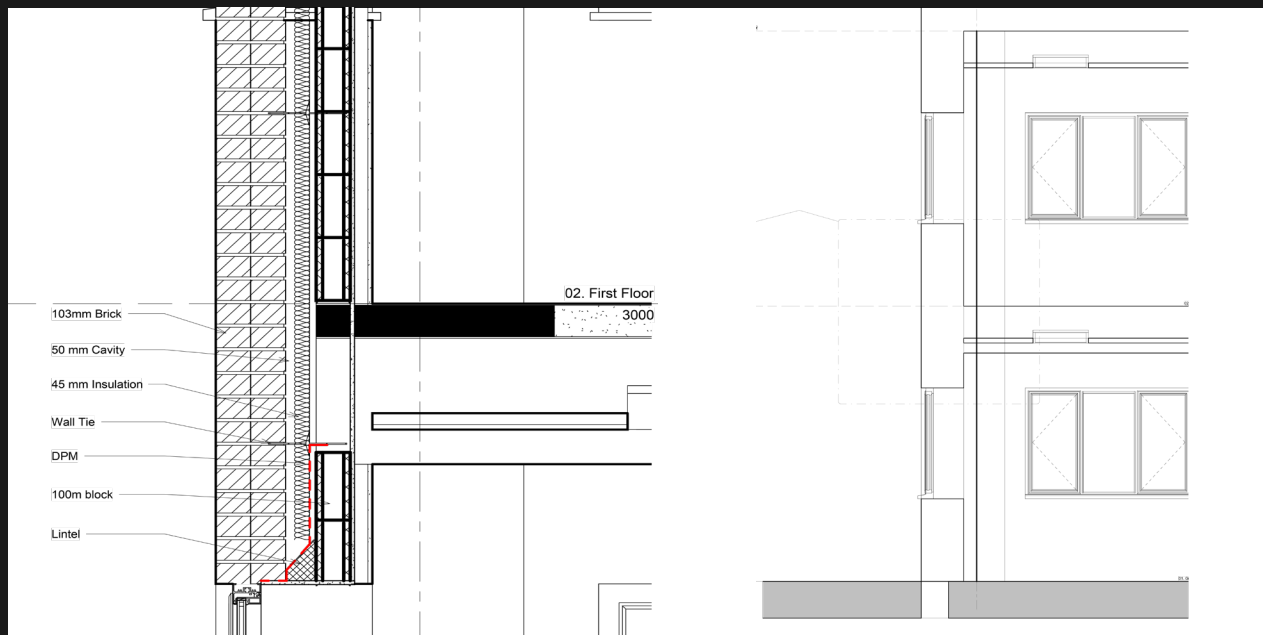


URBAN DESIGN

1
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The 905 Project

Volume I



BEHIND THE WALL

In architecture school I learned a lot about how walls are designed. In this particular wall, I left a 50 mm cavity in between the outer brick and the insulation to allow for natural expansion that happens to bricks due to weather.

I contemplated adding a rain screen and a wind screen to the insulation of the wall, but decided against it due to the buildings compact size. There were a few classes I took in school that helped me to calculate the depth and size walls depending on the building.

The windows used in this building are triple pane. It was important that I used the best quality materials. While its expensive to design a building that will last 100 years, the goal was to get as close as possible.

RESOURCES AND FINANCIAL INFO

Financial possibilities and costs.

01 This building was designed with the us of youtube tutorials:
<https://www.youtube.com/channel/UC17rnDjAlowh68LIGcaeL9A>

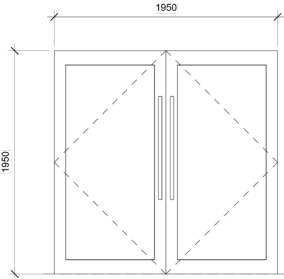
02 Hidden costs and unexpected issues will always arise in most projects, and planning for them early on will make the building/construction process much smoother.

03 Having said this, I estimate this project will cost \$196 per sqaure foot. This estimate comes from research done online:
<https://tinyurl.com/mwjmcx3x>

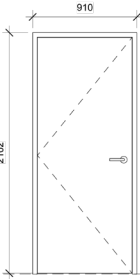
SWAMI

OFFICE BUILDING

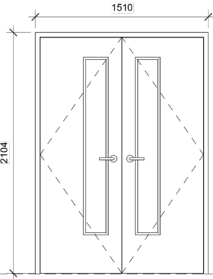
WINDOW AND DOOR SCHEDULE



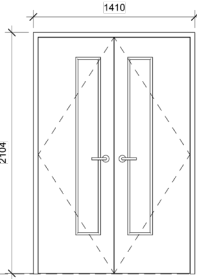
EXTERNAL DOOR TYPE: EDT-01
QTY: 04



TYPE: DT-01
QTY: 11



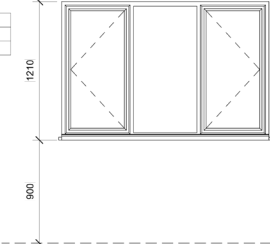
DOOR TYPE: DT-02
QTY: 04



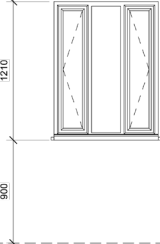
DOOR TYPE: DT-03
QTY: 02

DOOR SCHEDULE							
Door Type	Width	Height	Fire Rating	Description	Finish	Count	Comments
01. Ground Floor							
DT-01	910	2110	FD30S	Internal Single Door	Oak	7	
DT-03	1410	2110	FD30S	Internal Double Door	Oak	1	
DT-02	1510	2110	FD30S	Internal Double Door	Oak	2	
EDT-01	1950	2125		External Double Door	Aluminium	4	
02. First Floor							
DT-01	910	2110	FD30S	Internal Single Door	Oak	4	
DT-03	1410	2110	FD30S	Internal Double Door	Oak	1	
DT-02	1510	2110	FD30S	Internal Double Door	Oak	2	
Grand total: 21							

WINDOW SCHEDULE				
Window Type	Width	Height	Description	Comments
01. Ground Floor				
WT-02	910	1210	uPVC Triple Pane	
WT-01	1810	1210	uPVC Triple Pane	
02. First Floor				
WT-02	910	1210	uPVC Triple Pane	
WT-01	1810	1210	uPVC Triple Pane	
Grand total: 41				



WINDOW TYPE: WT-01
QTY: 21



WINDOW TYPE: WT-02
QTY: 22

No.	Description	Date

CODE

SUITABILITY DESCRIPTION

STATUS

PURPOSE OF ISSUE



AUTODESK
www.autodesk.com/revit

PROJECT

OFFICE BUILDING

TITLE

DOOR AND WINDOW LEGEND

CLIENT

AUTODESK

DRAWN BY
KY

CHECKED BY
KY

DATE
1999 Date

SCALE (B2)

PROJECT NUMBER

202

REV

DOORS AND WINDOWS

The best quality materials were used in this building. In order to ensure aesthetic value, oak is the material of choice.

In order to make sure the overall quality is at its highest level, every detail must be thought through at the highest level. No shortcuts taken. Only the best quality materials will be used. The furniture will be solely manufactured in Italy.

Interior design helps to uplift our spirits and also helps to effectively optimize the space in a building making the best use of the available space. While this project doesn't dive to much into interior design, I acknowledge the fact that interior design plays a huge role in creating the type of atmosphere I'm striving for.

SWAMI

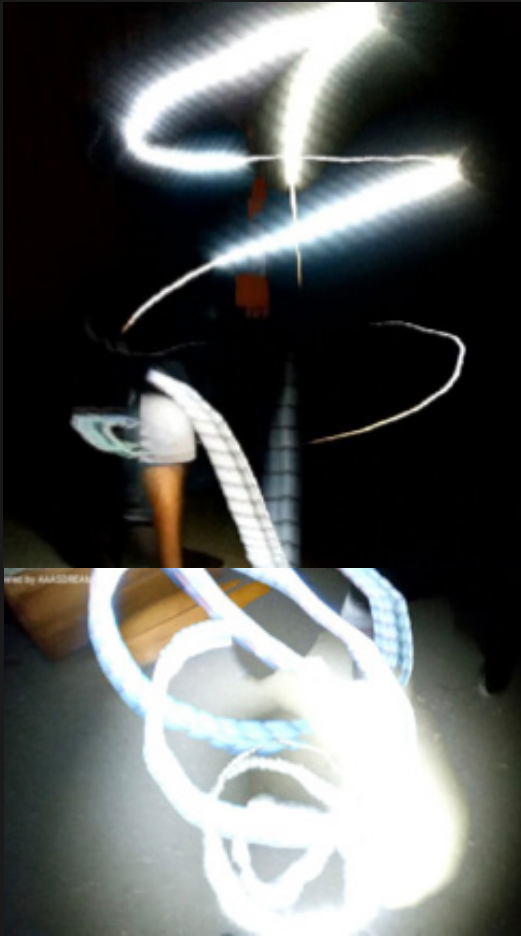
LIGHT SCULPTURE

FORM MAKING

02

ABOUT THE DESIGN

In this project we were asked to explore light as a method of composition. Using long exposure photography, I captured myself creating light drawings in the air with a flashlight. My aim was to create a spinning motion that I could recreate in an architectural model. The image on the right is my interpretation of the light drawings I created in a hand made model.



INSPIRATION

Jagged Design



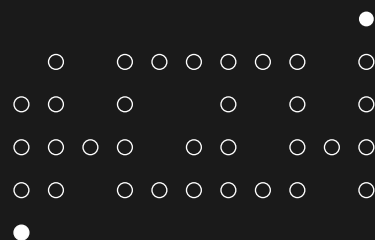
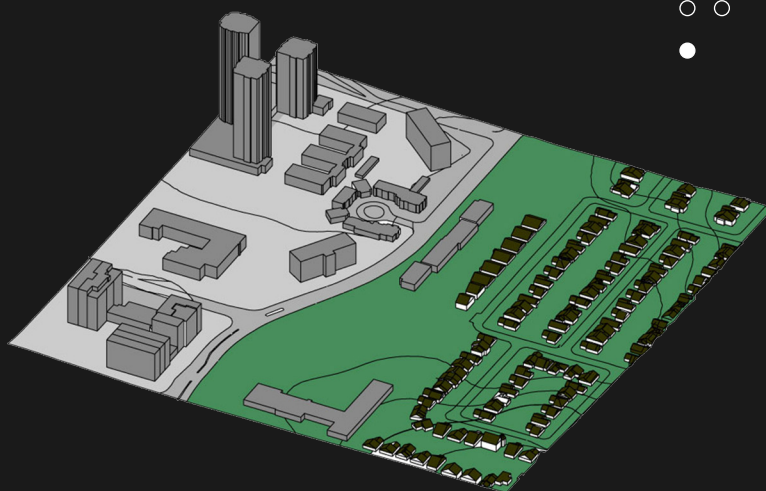
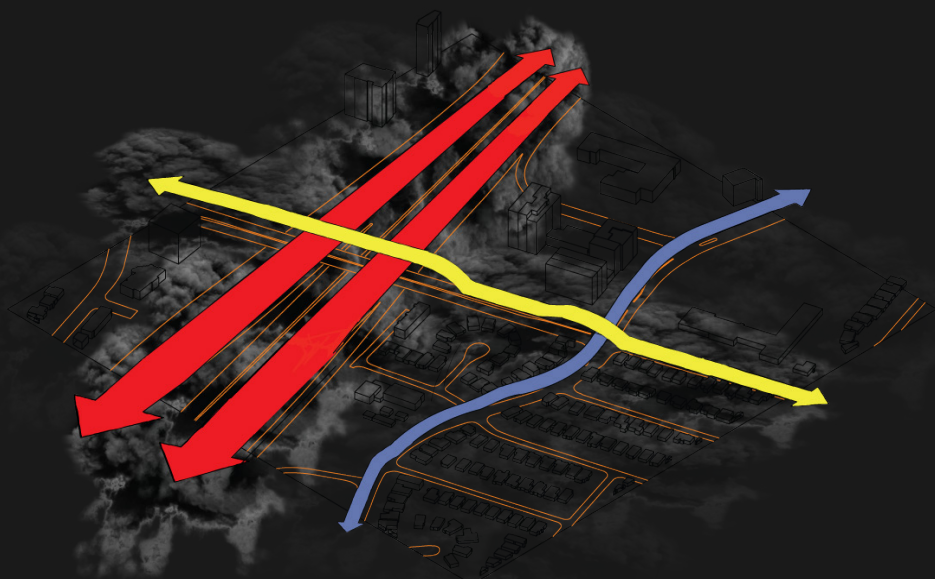
I've always been inspired by triangular design. Not only are triangles incredibly strong, when placed together like in the above design, it creates a jagged, sharp design. Some would say it looks 'evil' or 'dark', and that's why a curve was placed around it to round-out the design.

MAPPING AIR POLLUTION

As climate change becomes increasingly obvious in our current weather systems, we thought it might be interesting to look at the relationship between air pollution and roads.

In this class we were given a 60 by 60 km section of the Greater Toronto Area and asked to focus on a key aspect that makes our site unique. I started by analyzing the road systems on my map in order to look at how traffic moves on my site.

HOW MUCH AIR POLLUTION DO
AUTOMOBILES REALLY PRODUCE
IN SUBURBAN AREAS?



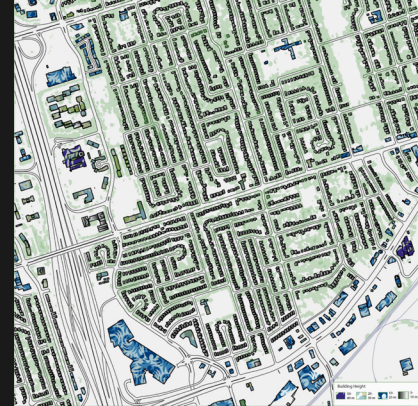
MAPPING

AIR POLLUTION

ABOUT THE PROJECT

I took panoramic pictures of traffic as a reference when creating hand drawings of the traffic density. Using the Canadian Climate Associations data and maps of air pollution, I proceeded to relate both the road circuits and the traffic together in order to create a visual understanding of how our cars are effecting climate change. The image on the right highlights the buildings on the site.

01



START A REUSE PROJECT

02



The image on the left shows the greenery that can be found on the 60km x 60km site I was given. The image on the right is a hand drawn sketch of the road systems, as I was getting acquainted with my site.

03



FINAL RESULT

01

As you can see on the right, the most traffic pollution is near the highway. Due to traffic density, that makes perfect sense.

02

I was also measuring how building height relates to air pollution. My hypothesis was that the bigger the building, the more energy it requires, and therefore more pollution is created. No significant conclusion was determined.

04



SWAMI

LIBRARY

BUILDING DESIGN

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1A PERSONAL
PROJECT

Library is a mixed-use study commons placed in a picturesque imagined landscape.

DESIGN

04

This "green" project includes 2,000 sq m of new building construction and creates an inviting library experience featuring a mix of study spaces, hang-out areas, food and beverage stations, and shelves to hold over 5000 books.

LOCATION	SIZE	YEAR	DESIGNER
Imagined	1118 sq. m	2021	SWAMI

OVERVIEW

PHASES OF DESIGN	Rough Sketch	Floor Plans	Detailing	Finished Project
	Form Shaping	Landscape Design	Steel framed building with curtain wall and glass ornamentation. Each floor has a study commons, kitchen and hang out areas. Landscape includes detailed terrain, road and water feature.	Views, rendering and sheet creation

The goal is to create a vibrant, walkable and sustainable library for students, the university community and neighbors from the surrounding region. While working on this project, my goal was to design a library form that both looks beautiful but can also exist in the real world. Glass has always been a fascinating material to me, and I enjoyed playing with different forms in pursuit of one that wraps around the building in a way that adds depth to the buildings original form.

GENERAL INFORMATION

LIBRARY

REAL ESTATE IN AN IMAGINARY PLACE.



'Library' aims to be the perfect modern library. Simple yet beautiful. Sleek yet traditional.

A significant challenge was doing this without guidance or instruction. This was my second time using REVIT, so I used resources from google and youtube to learn the basics of how to use the software.

'Library' is a five story steel framed structure with masonry and curtain wall windows. I wanted to create a modern library design that could be appreciated by the masses.

DETAILS

ROOMS

3 IT rooms
3 conference rooms
5 washrooms
3 kitchens

FURNITURE

54 study tables
6 conference tables
288 chairs

DIMENSIONS (per floor)

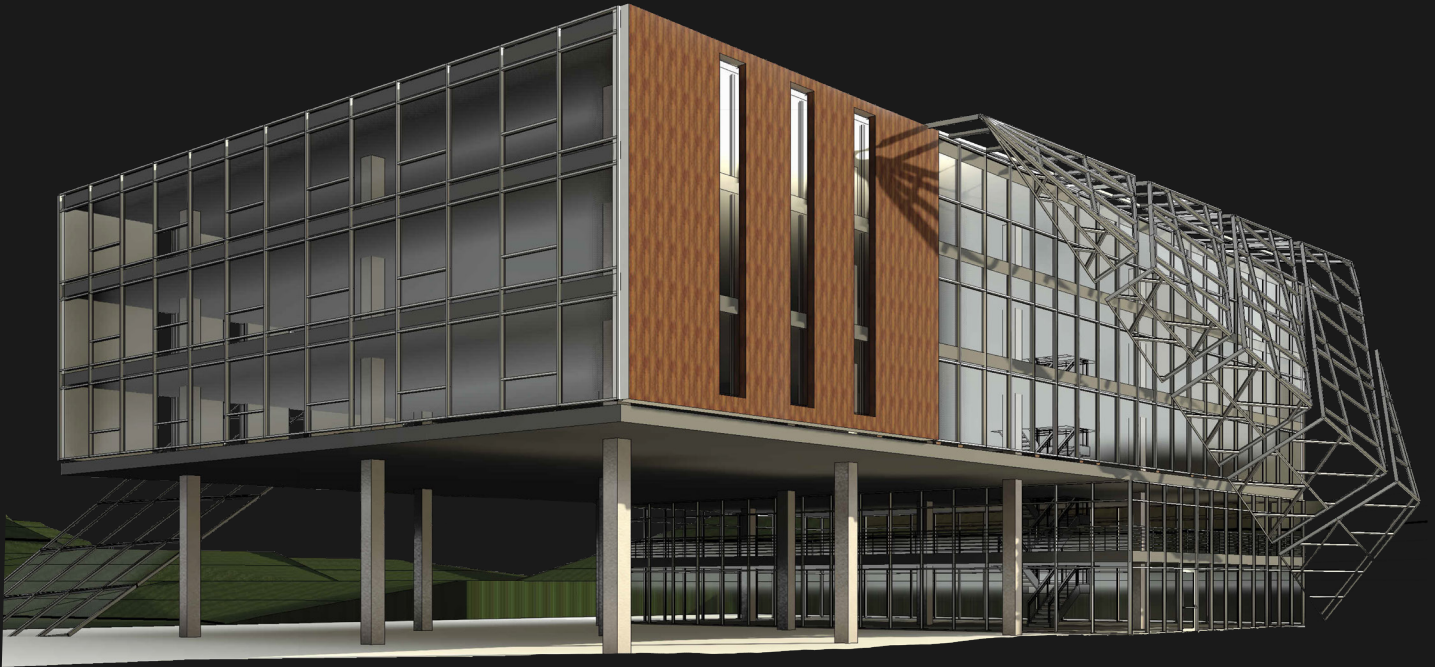
Building: 43 m x 26 m
Kitchen: 37 sq m
Conference Room: 82 sq m
Work Area: 878 sq m



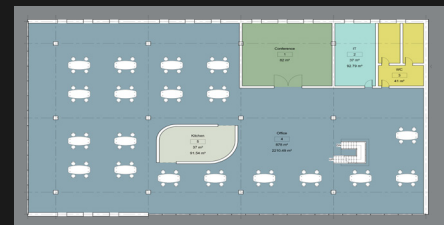
I decided to take this on as a personal project because I spent so much time studying in libraries over the past 4 years. I started thinking to myself, "how can I design the perfect library?"

1
6

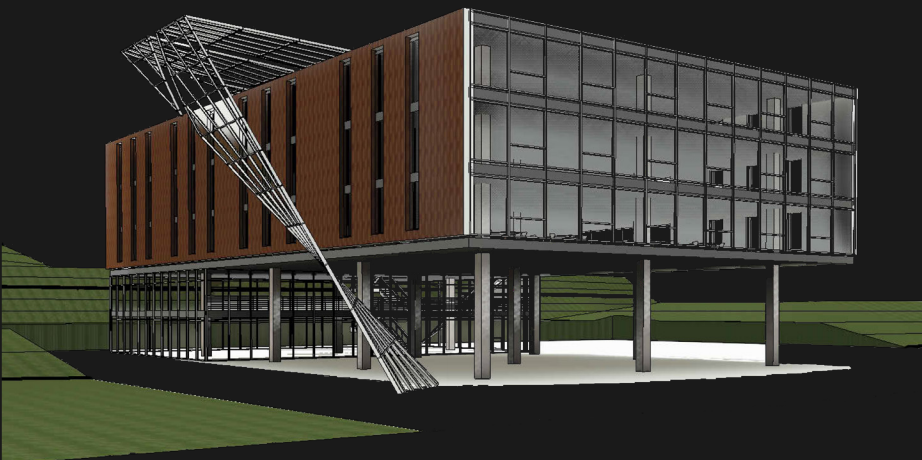
LIBRARY



FRONT PERSPECTIVE
Render



3rd Floor - Floor Plan



ANGLED PERSPECTIVE
Render

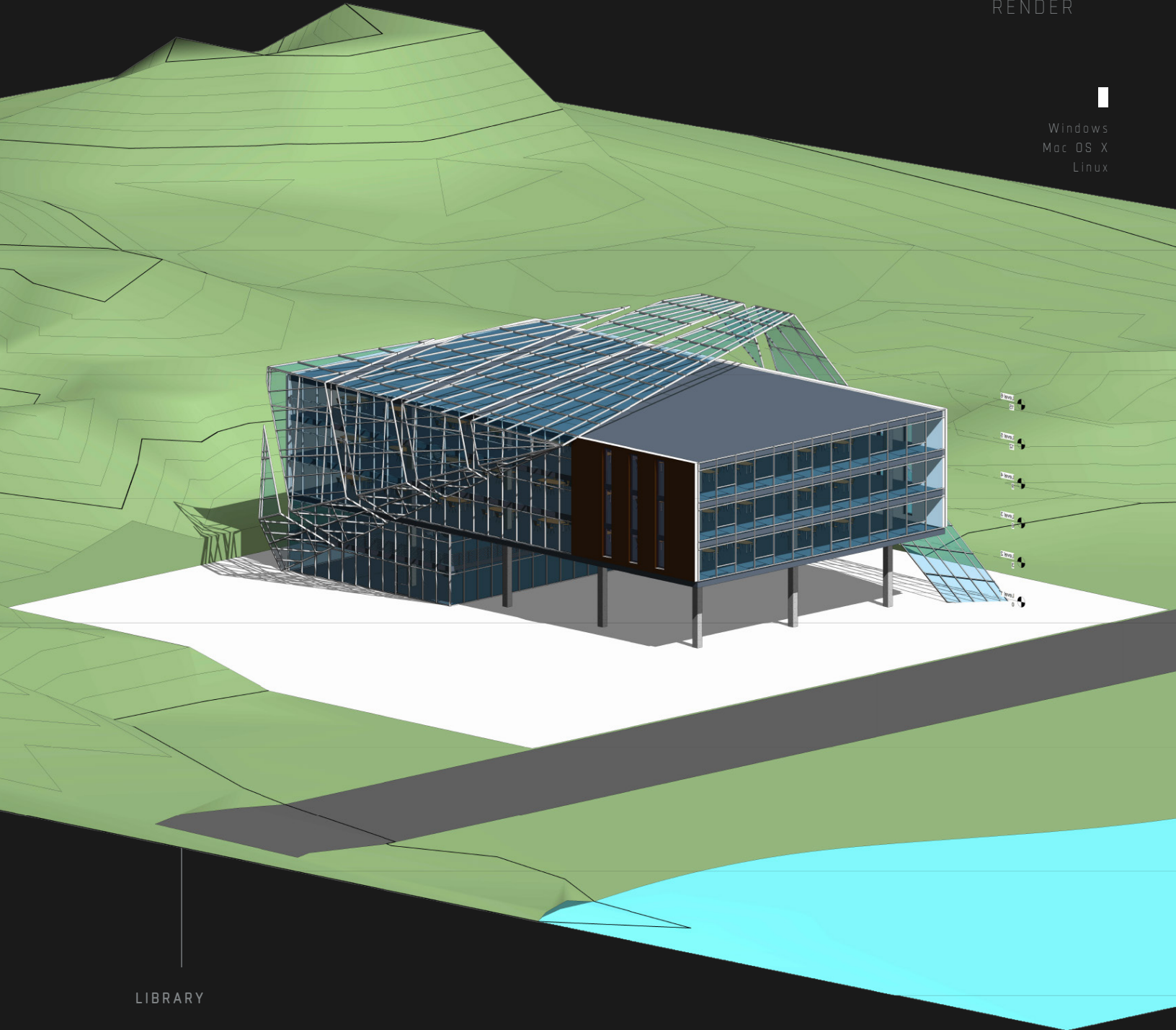
LIBRARY



LANDSCAPE
RENDER



Windows
Mac OS X
Linux



LIBRARY

1
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SWAMI

SWAMI

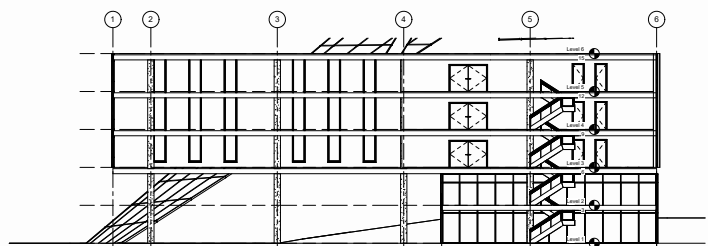
ELEVATIONS

1:100

WEST ELEVATION

EAST ELEVATION

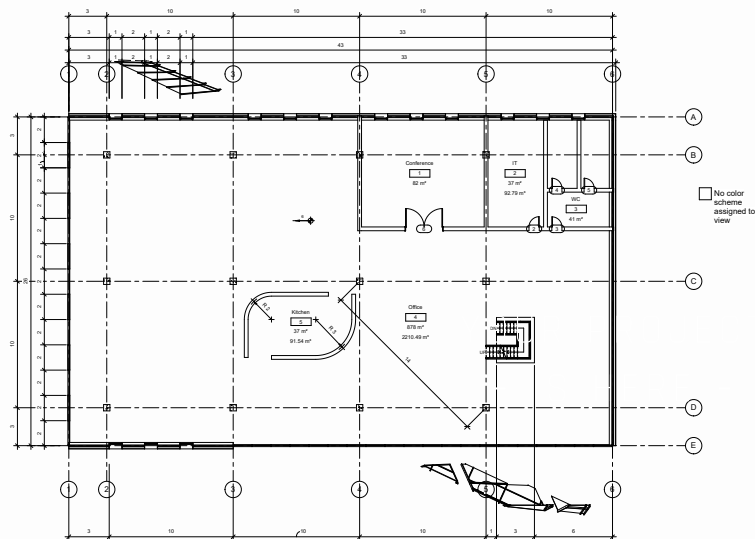
Elevations from either side of the structure highlight the beauty of the glass form that wraps around the building, without being obstructive to the viewers inside the building.



Section 1
1:100



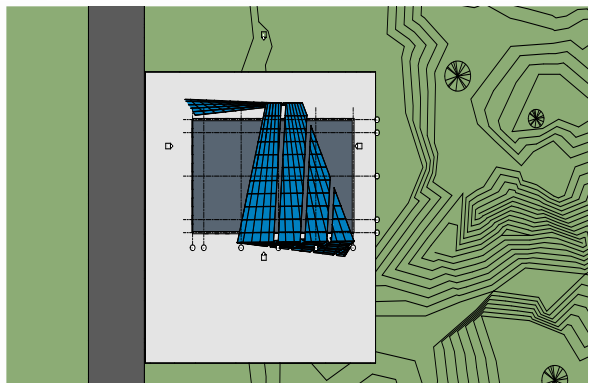
3D View 1



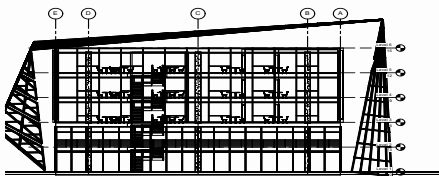
Level 3 with Rooms +
Dimensions Copy 1
1:100



Floor Plan



Site Plan



Elevation

05 FORM MAKING

Minus the three images in the middle, the forms on this page were created solely through code. These exercises helped me as a designer understand how programming can be used to generate new ideas.

CLASS

ARC465

DESIGNS

5

YEAR

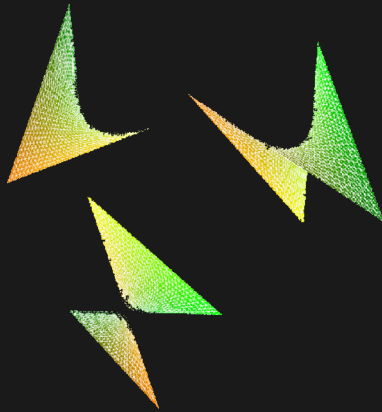
2019

TEAM

Wilson & Me

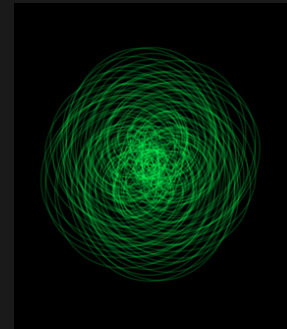
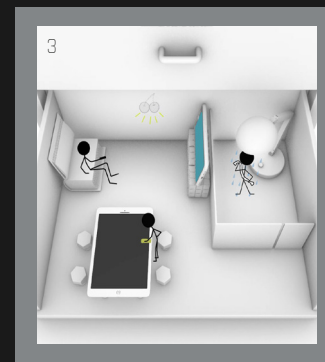
INSTRUCTOR

Andrew Bako

COMPUTATION
AND DESIGN

INFORMATION

The designs on the next page were created as part of a two part assignment series from a course called "ARC465 - Lazy Computing". The second phase builds off the first creating one massive project. The theme was 'recycled materials' and our objective was to repurpose unused materials we created in a previous design exercise.

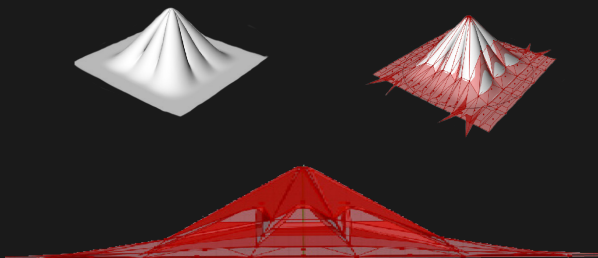
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9
DESIGN EXERCISES

INANIMATE OBJECTS

ABOUT THE ABOVE DESIGN

Building off the theme of 'recycled materials' as described above, the objective of this assignment was to use inanimate objects in a creative way that still resembles 'architecture'. We chose to use a side table as the enclosure, screws as the seats, an ipad as a table, a book as a couch, a phone as a TV, and a lamp as a shower.

THE PROCESS OF DEVELOPING THE FORMS ON THE NEXT PAGE BEGAN BY TRYING TO PIECE TOGETHER DIFFERENT PARTS OF AN ORIGINAL 'PILE OF TRASH' IN AN ATTEMPT TO SALVAGE SOMETHING



Please scan the QR code to watch a video animation on Instagram for the above mentioned project.



MND18

2
1

SWAMI

'PILE OF TRASH'

Screenshots from an animation video published to Instagram.

2

PHYSICAL
MODELS

3

FINAL
RENDERS

3

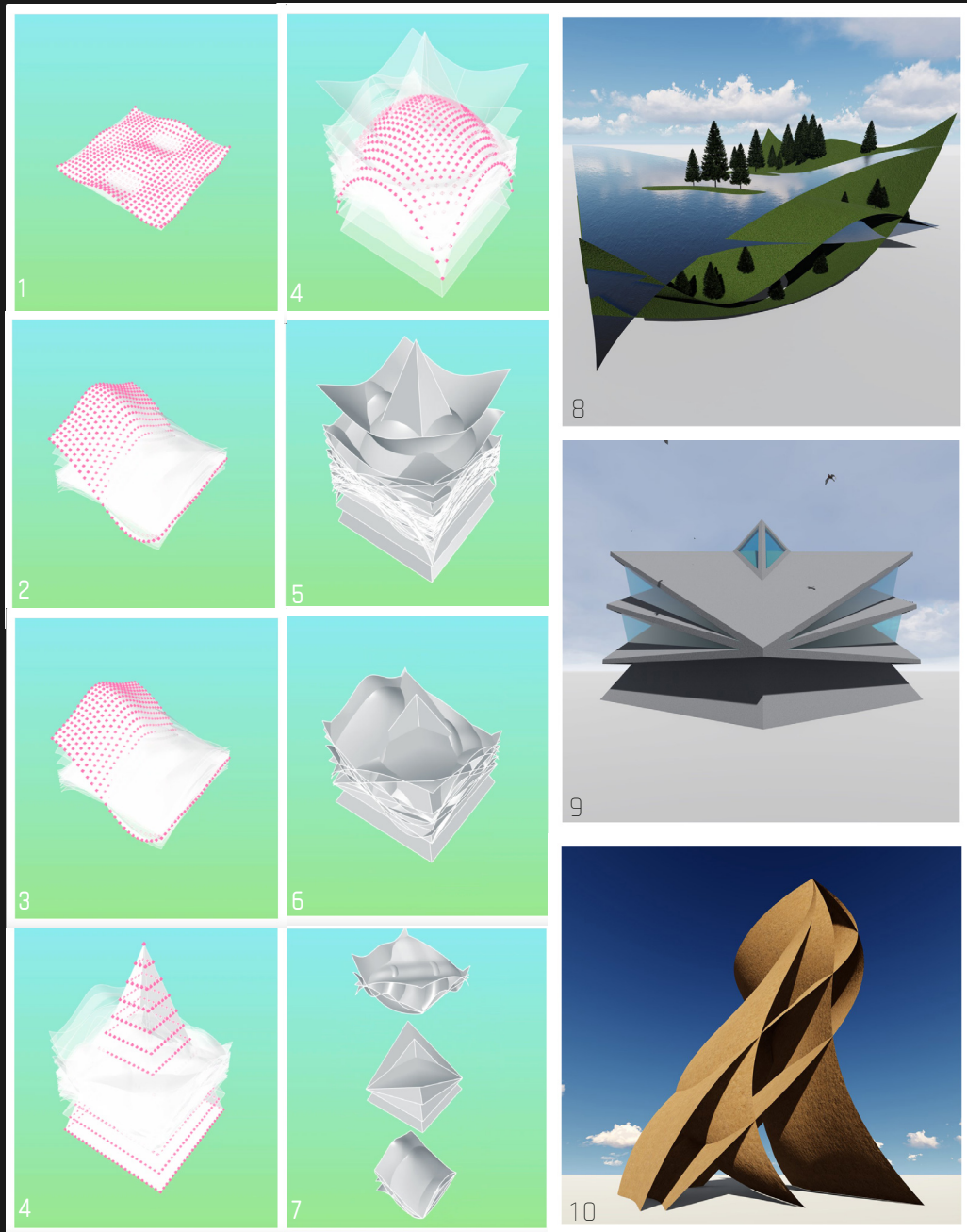
TOP
VIEWS

3

SECTION
DRAWINGS

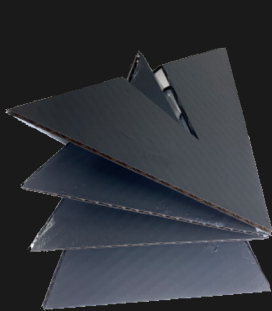
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ANIMATION
VIDEOS

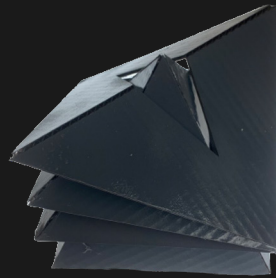




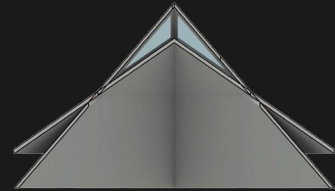
Please scan the QR code to watch a video animation on Instagram for the project shown in the past two pages



3D Model



3D Model



Section



3D Model



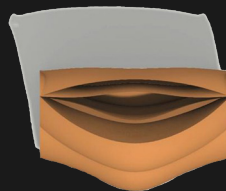
3D Model



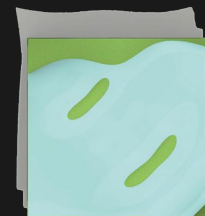
Section



Top View



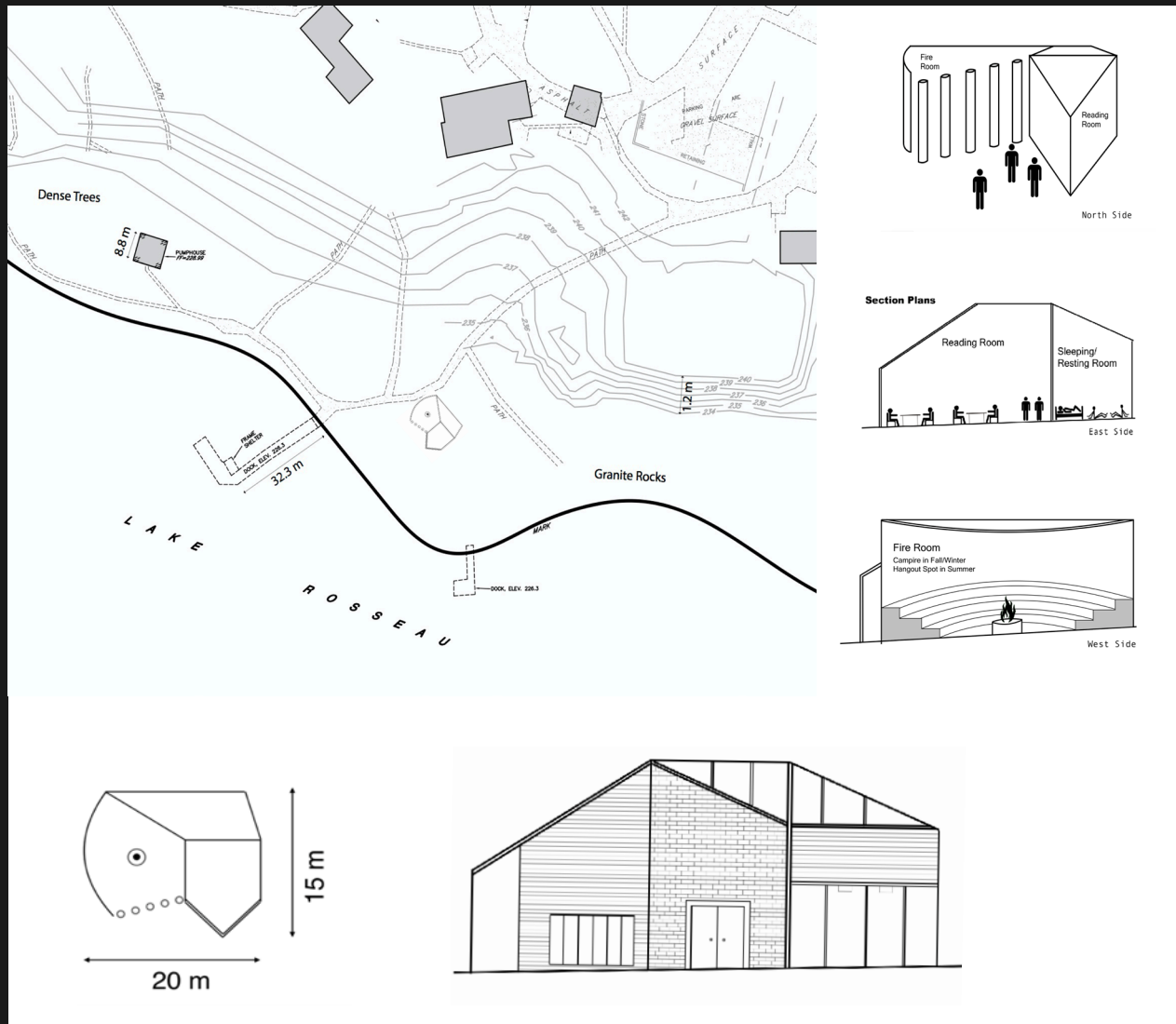
Top View



Top View

06 DESIGN COMPETITION

PAVILLION



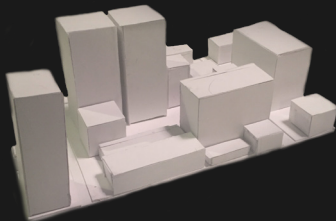
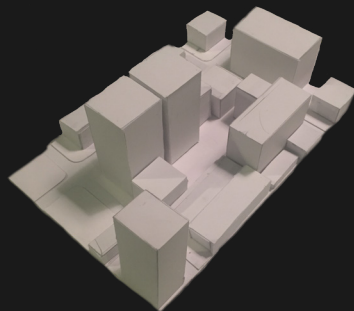
A private school in Muskoka Ontario challenged our class to design a pavilion that encompassed 3 different elements - a reading room, a resting room and a fire room. I decided to design a pavilion with a space for desks and chairs used for studying, and a space with bean bag chairs and carpeted ground for resting. The challenge was to interpret what exactly the school wanted for the fire room. I decided on a outdoor campfire space attached to the pavilion where students could regularly hang out and engage in social activities.

MODEL MAKING

SWAMI

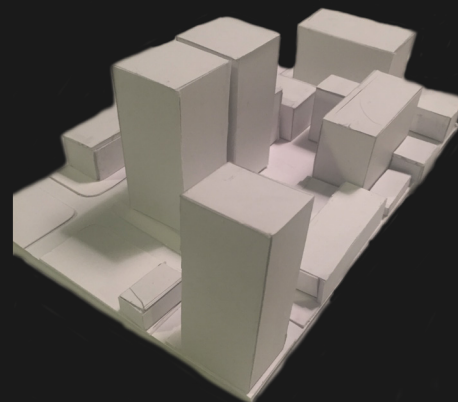
IN CONTEXT

In the right hand side images, my instruction was to take a monument, recreate it and recontextualize it. I chose a monument by Giuseppe Terragni.



The white coloured models were an exercise in form making. We were asked to place a series of prechosen buildings on a site in Toronto.

I decided upon placing buildings in a way that would allow for the easiest walking path. I also wanted to make sure no views were upstructed.



DESIGN EXERCISE

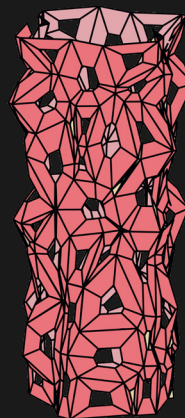
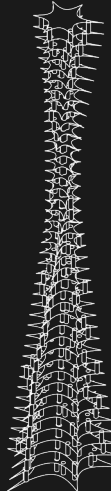
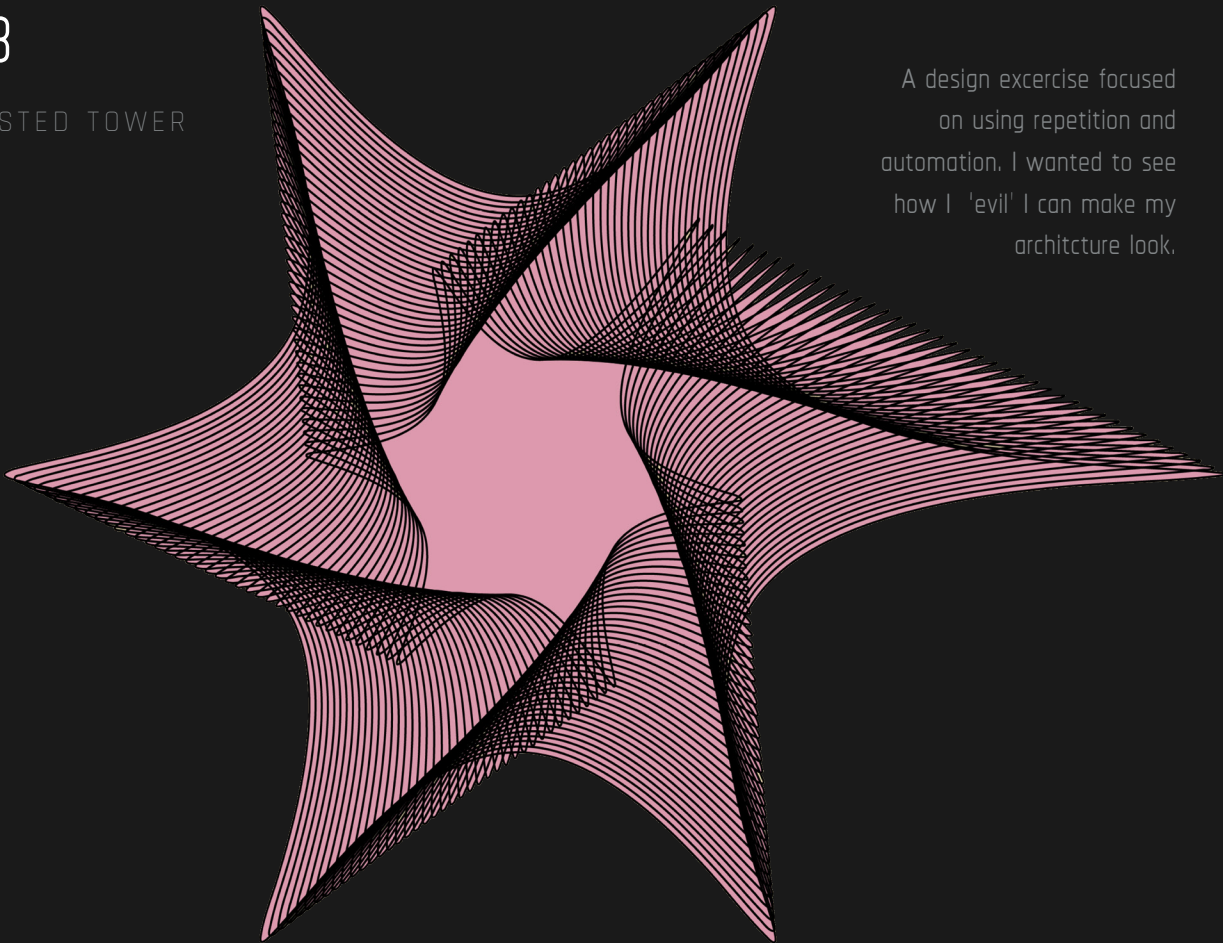
FORM MAKING

SWAMI

08

TWISTED TOWER

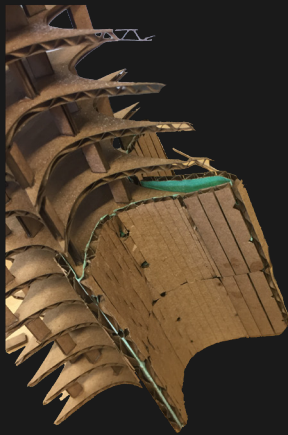
A design exercise focused on using repetition and automation. I wanted to see how I 'evil' I can make my architecture look.



TWISTED TOWER

REPETITION AND AUTOMATION

Through rigorous design classes, all students in the class delivered projects that represented the idea of a twisted tower.



INFORMATION

Through methods of 3D printing, laser cutting and hand made model making, over the course of 4 months I developed a concept for a twisting tower. The project required me to create a 40 story tower that used a shape for the floor plates and twisted and changed those floor plates in size as the tower grew.

In order to manage all the repetitive elements, most of the design was done in grasshopper.

I started by designing the tower, then adding structural columns, and finishing by adding a voronoi inspired facade. The challenge in this project wasn't as much designing the model as it was recreating my design into a physical version.



I laser cut each of the floor plates at a 1:200 scale and meticulously assembled the tower plate by plate. I tried two different facade's out to see which one I liked better and both were then 3D printed and placed on the model. This project gave me experience in model making, 3D design in Rhino, and programming in Grasshopper.

MODEL OF 3D FACADE



Explore a 3D model of the 3D printed facade



ARC200



2019

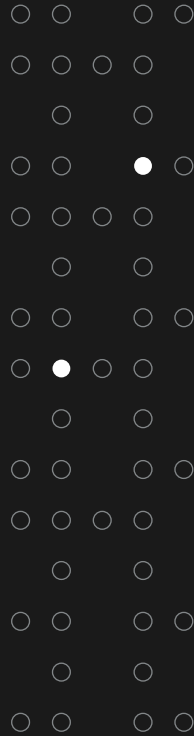
MOCK
UPS

SWAMI



DESIGN

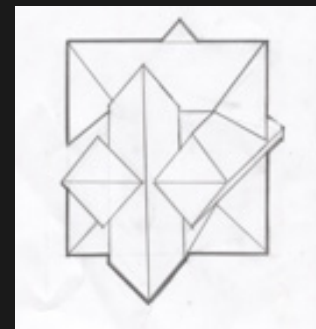
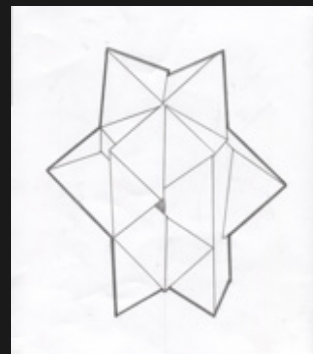
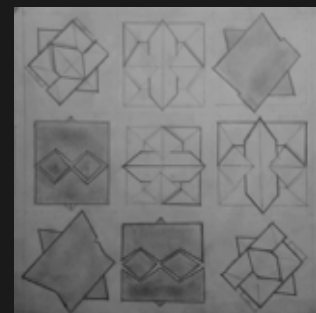
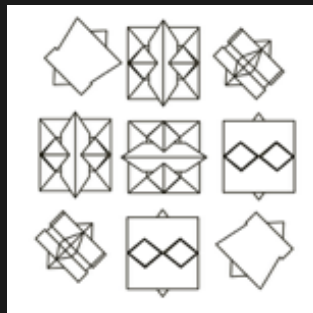
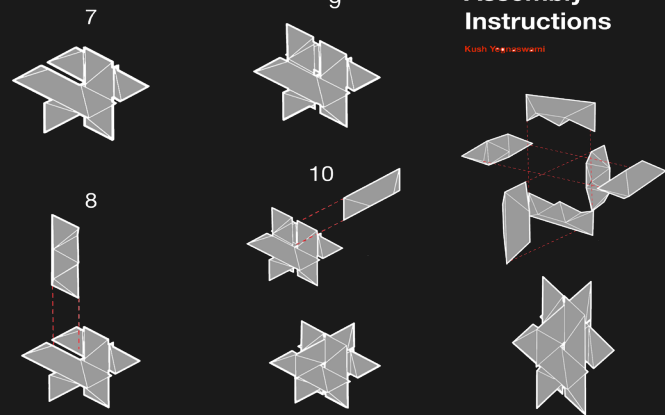
ARCHITECTURE ISN'T JUST BUILDINGS



Given a complex puzzle designed by Frank Lloyd Wright, my task was to draw and interpret the puzzle in various forms and create an instructional booklet on how to solve the puzzle. The toy I was given had 6 identical parts and came together to resemble a star shape.

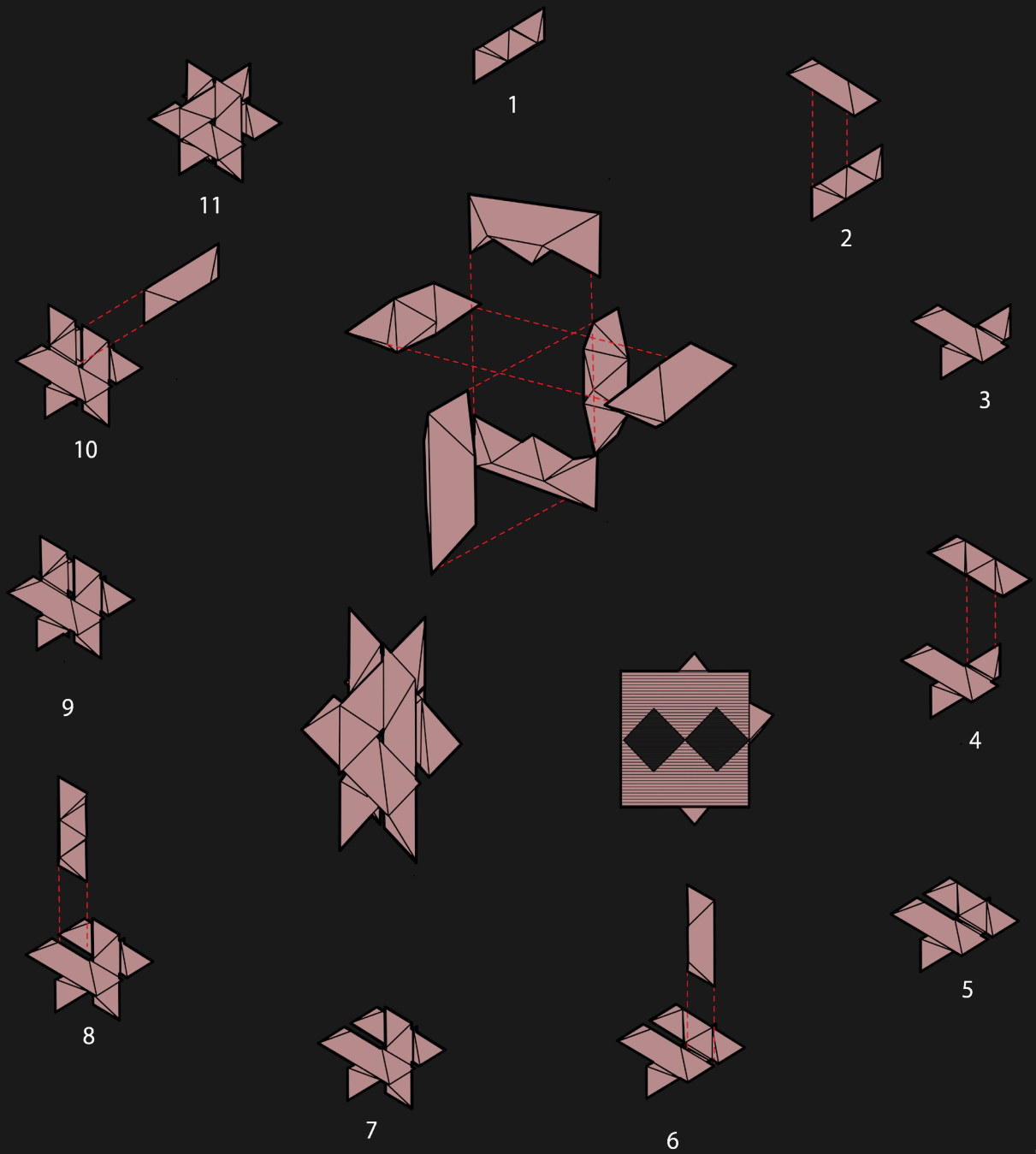
Each student was given a puzzle piece. We were asked to create a series of drawings for the puzzle piece, as well as create assembly instructions and a final drawing set.

It was tough to visualize it in axonometric and oblique views so I designed each of the 6 pieces in Rhino in order to help with the drafting and drawing of the toy. Throughout the course we drew elevations, sections, 2D views as well as isometric views from all sides. All these hand drawings were then recreated digitally and were finally put together in an instructional booklet for one to use in order to solve the puzzle.



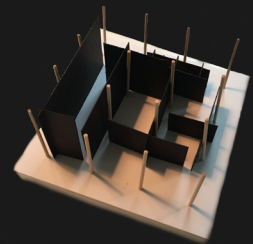
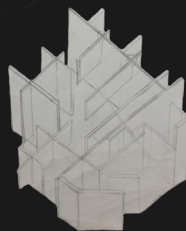
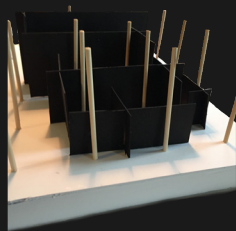
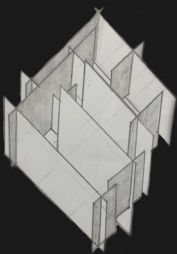
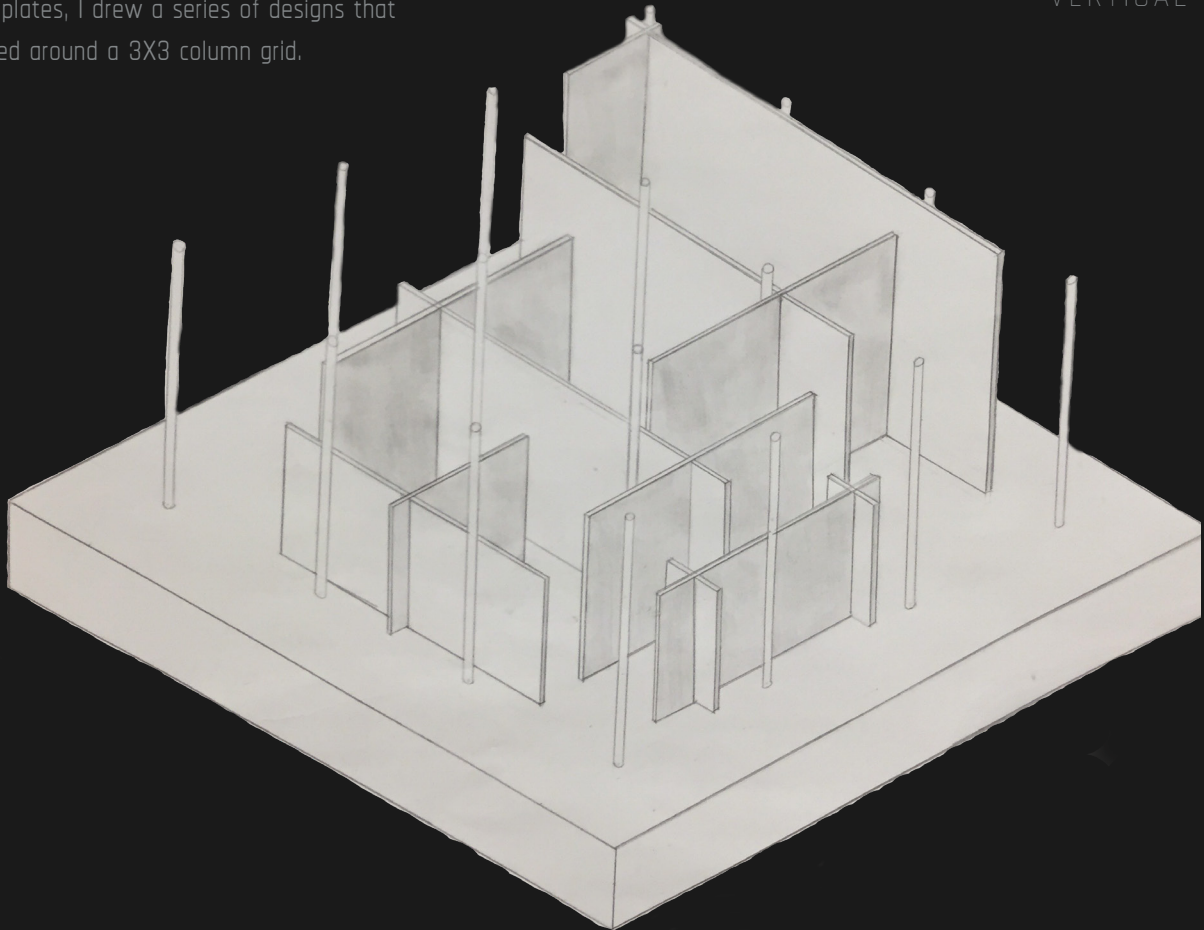
	ROUGH SKETCHING	PLANNING INSTRUCTIONS	COLOURING AND LINE-WEIGHTS
PLAN	FINAL DRAWINGS	DESIGN IN ILLUSTRATOR	

FINAL PRESENTATION



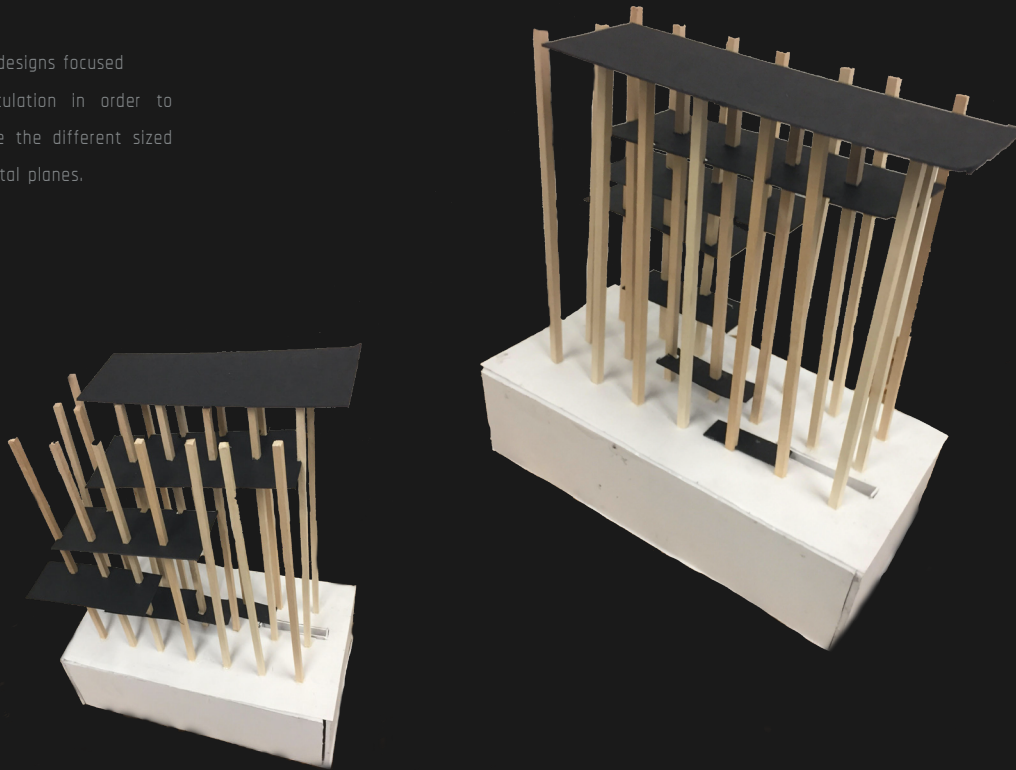
Only allowed to use vertical and horizontal floor plates, I drew a series of designs that worked around a 3X3 column grid.

HORIZONTAL VS
VERTICAL

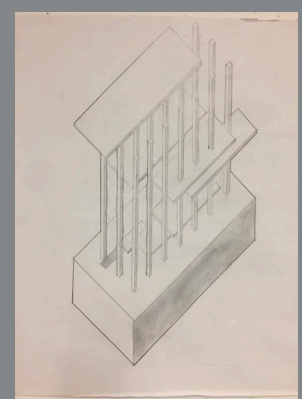
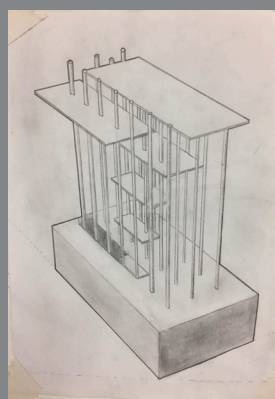
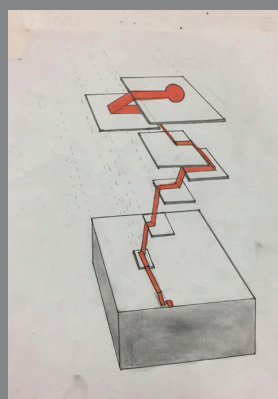
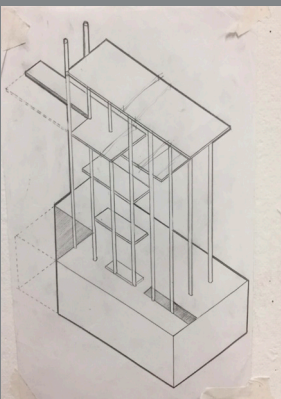




These designs focused
on circulation in order to
manage the different sized
horizontal planes.



I was able to
push my drafting
and drawing skills
in this course by
trying to
remain accurate
and precise with
my drawings.



I ENROLLED IN ARCHITECTURE SCHOOL WITHOUT BEING ABLE TO DRAW

I joined architecture school because I loved designing things. Whether it's a new sneaker concept, a car concept, or an interior design concept, being creative is always something that has interested me. But I never designed with a paper and pencil. That didn't feel natural. I preferred Photoshop and Microsoft Paint. When I got to architecture school, they actually taught me how to draw. A lot of people think you need to be able to draw in order to be an architect. But one of the most interesting things I learned was that actually the best architects aren't great at drawing either. Frank Gehry is notorious for his 'scribble' sketches.

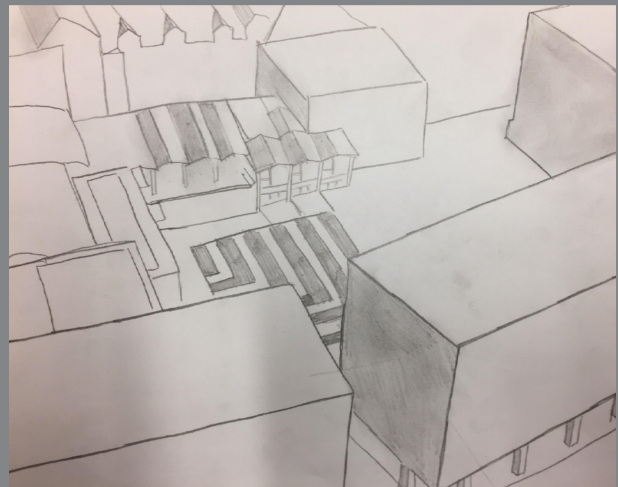
STUDIO SKETCHING

Our exercise was to draw a space that's meaningful to us. I chose to draw a studio space that I worked in day in, day out for my entire university career. Every day I would go to this studio and work on music. I lived in the curved building just opposite to it for the first year of Uni. This is the place where I became a master of my craft.

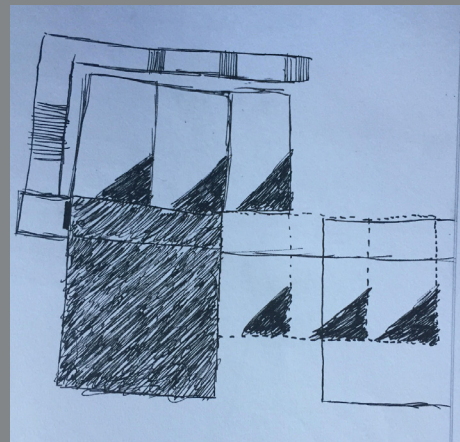
Inside View



Perspective



Rough Plan View



The purpose of this sketch is to be messy. My instructor wanted me to understand architecture is not about being perfect. It's about experimentation. And experimentation is never perfect.

SWAMI

FORM MAKING 2
0
1
8

12

MORE MODELLING

E X P E R I M E N T I N G
C U R V E S
W I T H



3
3

The 905 Project

FORM
MAKING

OVERVIEW

EXPERIMENTING WITH
CURVES

12

Building off the column grid presented on page 30 and 31, this project presents how curved architecture might wrap around the columns.



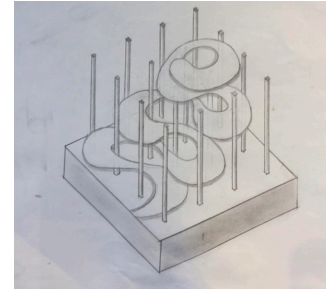
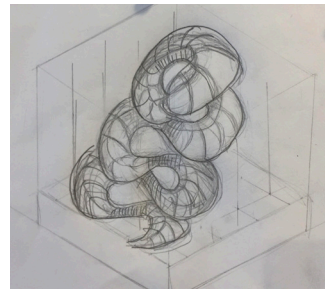
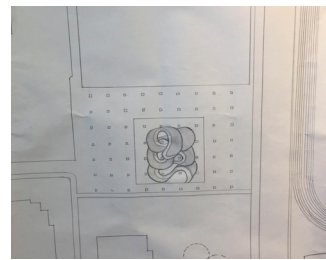
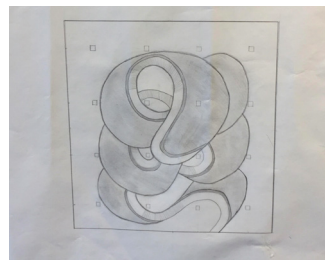
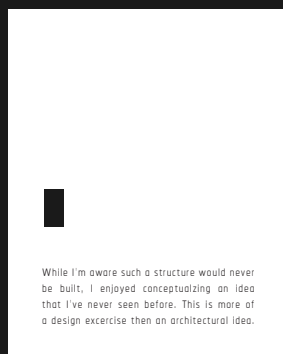
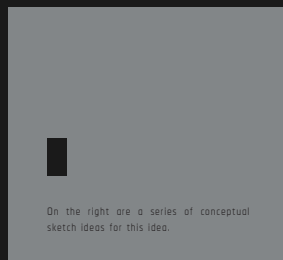
The idea is to have steel curved beams as reinforcement, a walk pathway within the structure, and completely wooden exterior.



INFORMATION

I had a weird and wonky idea that I needed to get out of my system. I was interested in seeing how practical an idea for a snake-like, curvy piece of architecture would be. My idea was a long and tubular design that wrapped around the column grids provided. When it came down to modelling my ambitious idea, I found that it would be much more effective to model just one part of the design due to its repetitive nature rather than trying to model the whole thing. I can admit that my design was unsuccessful in an attempt to be practical, but I found my project successful in the sense that I was able to fully think through and model a design idea.

CURVES

1
6



SWAMI